

Assignment-1

Subject -Computer Organization & Architecture

Session-2026-27

Subject Code-BCS-302

Section-C/J

Q1.A digital computer has a common bus system for 16 registers of 32 bits each. The bus is conducted with multiplexers.

- How many selection inputs are there in each multiplexer
- What sizes of multiplexers are needed?
- How many multiplexers are there in the bus?

Q2. Specify the Control word that must be applied to the processor to implement the following micro-operation

$R1 \leftarrow R2 + R3,$

$R4 \leftarrow R4$

$R5 \leftarrow R5 + 1$

$R6 \leftarrow \text{shl } R1$

$R7 \leftarrow \text{input}$

Q3-Explain functional units of computer system in detail.

Q4-What are the different types of Buses used in computer architecture?

Q5- What is the term BUS arbitration meant? Why is it needed? How can bus arbitration be implemented in Daisy chaining scheme?

Q6-Write a program to evaluate the following arithmetic expression:

$$X = (A - B + C * (D * E - F)) / (G + H * K)$$

(i) Using general register computer with 3-address instruction

(ii) Using an accumulator type computer with one address instruction